UNIVERZITET U NOVOM SADU
FAKULTET TEHNIČKIH NAUKA
KATEDRA ZA AUTOMATIKU I UPRAVLJANJE SISTEMIMA

Linearan matematički model Primena Laplasove transformacije (deo B)

Modeliranje i simulacija sistema

Upravljanje, modelovanje i simulacija sistema

Inverzna Laplasova transformacija

$$F(s) = \frac{P(s)}{Q(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \ldots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \ldots + a_1 s + a_0}$$

Postupak:

1. Rastavljanje $F(s)$ na parcijalne sabirke
2. Prepoznavanje sabiraka u tablicama Laplasove transformacije i pisanje originala

Od posebnog značaja su **polovi** funkcije $F(s)$, tj. koreni $Q(s) = 0$, i tu se mogu uočiti četiri karakteristična slučaja:

- Svi polovi funkcije $F(s)$ su realni i prosti
- Funkcija $F(s)$ ima višestruke realne korene
- Postoje konjugovano-kompleksni polovi, a realni su, ako postoje, prosti
- Funkcija $F(s)$ ima višestruke konjugovano kompleksne polove

Polovi funkcije su realni i jednostruki

$$F(s)=\frac{P(s)}{Q(s)}=\frac{P(s)}{(s-s_1)(s-s_2)\dots(s-s_n)}=\frac{K_1}{s-s_1}+\frac{K_2}{s-s_2}+\ldots+\frac{K_n}{s-s_n}=\sum_{k=1}^{n}\frac{K_k}{s-s_k}$$

$$(s-s_1)\frac{P(s)}{Q(s)}=\frac{P(s)}{(s-s_2)\dots(s-s_n)}=K_1+(s-s_1)\frac{K_2}{s-s_2}+\ldots+(s-s_1)\frac{K_n}{s-s_n}$$

$$\frac{P(s_1)}{(s_1-s_2)\dots(s_1-s_n)}=K_1$$

Uopšteno:

$$K_k=\left[(s-s_k)\frac{P(s)}{Q(s)}\right]_{s=s_k},\quad k=1,2,\dots,n$$

$$K_k=\lim_{s\to s_k}\left[(s-s_k)\frac{P(s)}{Q(s)}\right]=\lim_{s\to s_k}\frac{\frac{d}{ds}(s-s_k)P(s)}{\frac{d}{ds}Q(s)}=\frac{P(s_k)}{Q'(s_k)}$$

Original:

$$f(t)=\mathcal{L}^{-1}\left\{\sum_{k=1}^{n}\frac{K_k}{s-s_k}\right\}=\sum_{k=1}^{n}K_k\cdot e^{s_k t},\quad t>0$$

Neki polovi funkcije su realni višestruki

$$F(s) = \frac{P(s)}{Q(s)} = \frac{P(s)}{(s-s_1)^3(s-s_4)\ldots(s-s_n)} = \frac{K_{11}}{(s-s_1)^3} + \frac{K_{12}}{(s-s_1)^2} + \frac{K_{13}}{(s-s_1)} + \sum_{k=4}^{n} \frac{K_k}{s-s_k}$$

Ovo je primer sa trostrukim realnim polom.

$$(s-s_1)^3 \frac{P(s)}{Q(s)} = K_{11} + (s-s_1)K_{12} + (s-s_1)^2 K_{13} + (s-s_1)^3 \sum_{k=4}^{n} \frac{K_k}{s-s_k}$$

$$K_{11} = \left[(s-s_1)^3 \frac{P(s)}{Q(s)}\right]_{s=s_1}$$

$$K_{12} = \left[\frac{d}{ds}(s-s_1)^3 \frac{P(s)}{Q(s)}\right]_{s=s_1}$$

$$K_{13} = \frac{1}{2}\left[\frac{d^2}{ds^2}(s-s_1)^3 \frac{P(s)}{Q(s)}\right]_{s=s_1} \qquad K_{rm} = \frac{1}{(m-1)!}\left[\frac{d^{m-1}}{ds^{m-1}}(s-s_r)^p \frac{P(s)}{Q(s)}\right]_{s=s_r}, m = 1,2,\ldots p$$

$$f(t) = \frac{K_{11}}{2} t^2 e^{s_1 t} + K_{12} t e^{s_1 t} + K_{13} e^{s_1 t} + \sum_{k=4}^{n} K_k \cdot e^{s_k t}, \quad t > 0$$

Polovi funkcije su konjugovano kompleksni

$$F(s)=\frac{P(s)}{Q(s)}=\frac{P(s)}{(s-s_1)(s-s_1^*)(s-s_3)\ldots(s-s_n)}=\frac{K_1}{s-s_1}+\frac{K_1^*}{s-s_1^*}+\frac{K_3}{s-s_3}+\ldots+\frac{K_n}{s-s_n}$$

$$s_1=-\alpha+j\omega,\qquad s_1^*=-\alpha-j\omega$$

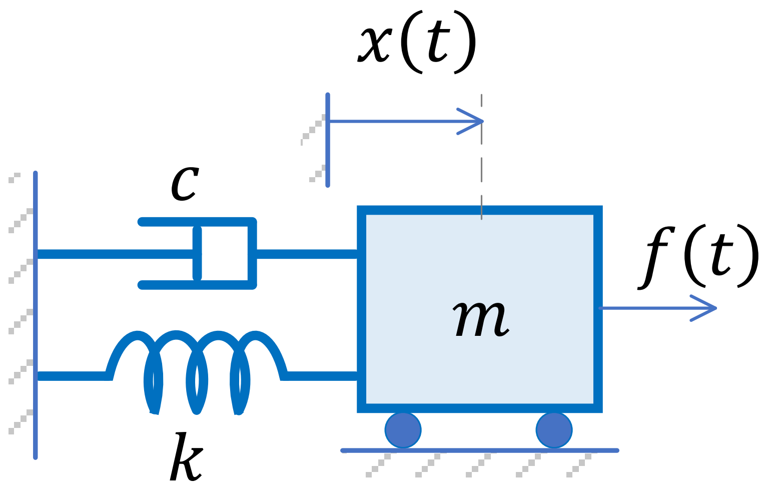
$$K_1=a+jb,\qquad K_1^*=a-jb$$

$$K_1=a+jb=\left.\frac{P(s)}{Q'(s)}\right|_{s=-\alpha+j\omega}$$

$$F(s)=\frac{a+jb}{(s+\alpha)-j\omega}+\frac{a-jb}{(s+\alpha)+j\omega}+\sum_{k=3}^{n}\frac{K_k}{s-s_k}=\frac{2a(s+\alpha)-2b\omega}{(s+\alpha)^2+\omega^2}+\sum_{k=3}^{n}\frac{K_k}{s-s_k}$$

$$f(t)=2a\cdot e^{-\alpha t}\cos\omega t-2b\cdot e^{-\alpha t}\sin\omega t+\sum_{k=3}^{n}K_k e^{s_k t},\qquad t>0$$

Primer primene Laplasove transformacije



- Mehanički sistem je opisan diferencijalnom jednačinom

$$m\frac{d^2x(t)}{dt^2} + c\frac{dx(t)}{dt} + k \cdot x(t) = f(t)$$

$$f(t) = 0, \quad x(0) = x_0 = 1, \quad \frac{dx}{dt}(0) = 0$$

Primer – nastavak

- Primena Laplasove transformacije na dif. jedn. daje:

$$m\left(s^2X(s) - sx(0) - \frac{dx}{dt}(0)\right) + c(sX(s) - x(0)) + k \cdot X(s) = 0$$

- a nakon sređivanja:

$$(ms^2 + cs + k)X(s) = (ms + c)x(0)$$
$$X(s) = \frac{ms + c}{ms^2 + cs + k}x(0) = \frac{ms + c}{ms^2 + cs + k}$$

$$X(s) = \frac{P(s)}{Q(s)} = \frac{s + \frac{c}{m}}{s^2 + \frac{c}{m}s + \frac{k}{m}}$$

Primer – realni jednostruki polovi

- Za $k/m = 2$ i $c/m = 3$
Razvoj u sumu parcijalnih sabiraka

$$X(s) = \frac{s+3}{s^2+3s+2} = \frac{s+3}{(s+1)(s+2)} = \frac{a}{s+1} + \frac{b}{s+2} = \frac{2}{s+1} - \frac{1}{s+2}$$

- Original $x(t)$ se dobija promenim inverzne Laplasove transformacije (upotrebom tablica)

$$x(t) = \mathcal{L}^{-1}\{X(s)\} = \mathcal{L}^{-1}\left\{\frac{2}{s+1}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} = 2e^{-t} - e^{-2t}$$

Određivanje koeficijenata:

$$K_k = \left[(s - s_k)\frac{P(s)}{Q(s)}\right]_{s=s_k}$$

ili

$$K_k = \frac{P(s_k)}{Q'(s_k)}$$

$$a = \left[(s+1)\frac{s+3}{(s+1)(s+2)}\right]_{s=-1} = \left[\frac{s+3}{s+2}\right]_{s=-1} = 2$$

$$b = \left[(s+2)\frac{s+3}{(s+1)(s+2)}\right]_{s=-2} = \left[\frac{s+3}{s+1}\right]_{s=-2} = -1$$

$$Q'(s) = 2s + 3$$

$$a = \frac{P(-1)}{Q'(-1)} = \frac{2}{1}$$

$$b = \frac{P(-2)}{Q'(-2)} = \frac{1}{-1}$$

Primer – realni višestruki polovi

- Za $k/m = 4$ i $c/m = 4$ Razvoj u sumu parcijalnih sabiraka

$$X(s) = \frac{s+4}{s^2+4s+4} = \frac{s+4}{(s+2)^2} = \frac{a}{s+2} + \frac{b}{(s+2)^2} = \frac{1}{s+2} + \frac{2}{(s+2)^2}$$

- Original x(t) se dobija primenom inverzne Laplasove transformacije (upotrebom tablica)

$$x(t) = \mathcal{L}^{-1}\{X(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} + \mathcal{L}^{-1}\left\{\frac{2}{(s+2)^2}\right\} = e^{-2t} + 2te^{-2t}$$

Određivanje koeficijenata:

$$b = \left[(s-s_1)^2\frac{P(s)}{Q(s)}\right]_{s=s_1} = \left[(s+2)^2\frac{s+4}{(s+2)^2}\right]_{s=-2} = [s+4]_{s=-2} = 2$$

$$a = \left[\frac{d}{ds}(s-s_1)^2\frac{P(s)}{Q(s)}\right]_{s=s_1} = \left[\frac{d}{ds}(s+2)^2\frac{s+4}{(s+2)^2}\right]_{s=-2} = 1$$

Primer – konjugovano-kompleksni polovi

- Za $k/m = 3$ i $c/m = 2$ Razvoj u sumu parcijalnih sabiraka

$$X(s) = \frac{s+2}{s^2+2s+3} = \frac{s+1+1}{(s+1)^2+(\sqrt{2})^2} = \frac{s+1}{(s+1)^2+(\sqrt{2})^2} + \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{(s+1)^2+(\sqrt{2})^2}$$

- Original x(t) se dobija promenim inverzne Laplasove transformacije (upotrebom tablica)

$$x(t) = \mathcal{L}^{-1}\{X(s)\} = \mathcal{L}^{-1}\left\{\frac{s+1}{(s+1)^2+(\sqrt{2})^2}\right\} + \frac{1}{\sqrt{2}} \cdot \mathcal{L}^{-1}\left\{\frac{\sqrt{2}}{(s+1)^2+(\sqrt{2})^2}\right\}$$

$$x(t) = e^{-t}\cos(t\sqrt{2}) + \frac{1}{\sqrt{2}}e^{-t}\sin(t\sqrt{2})$$

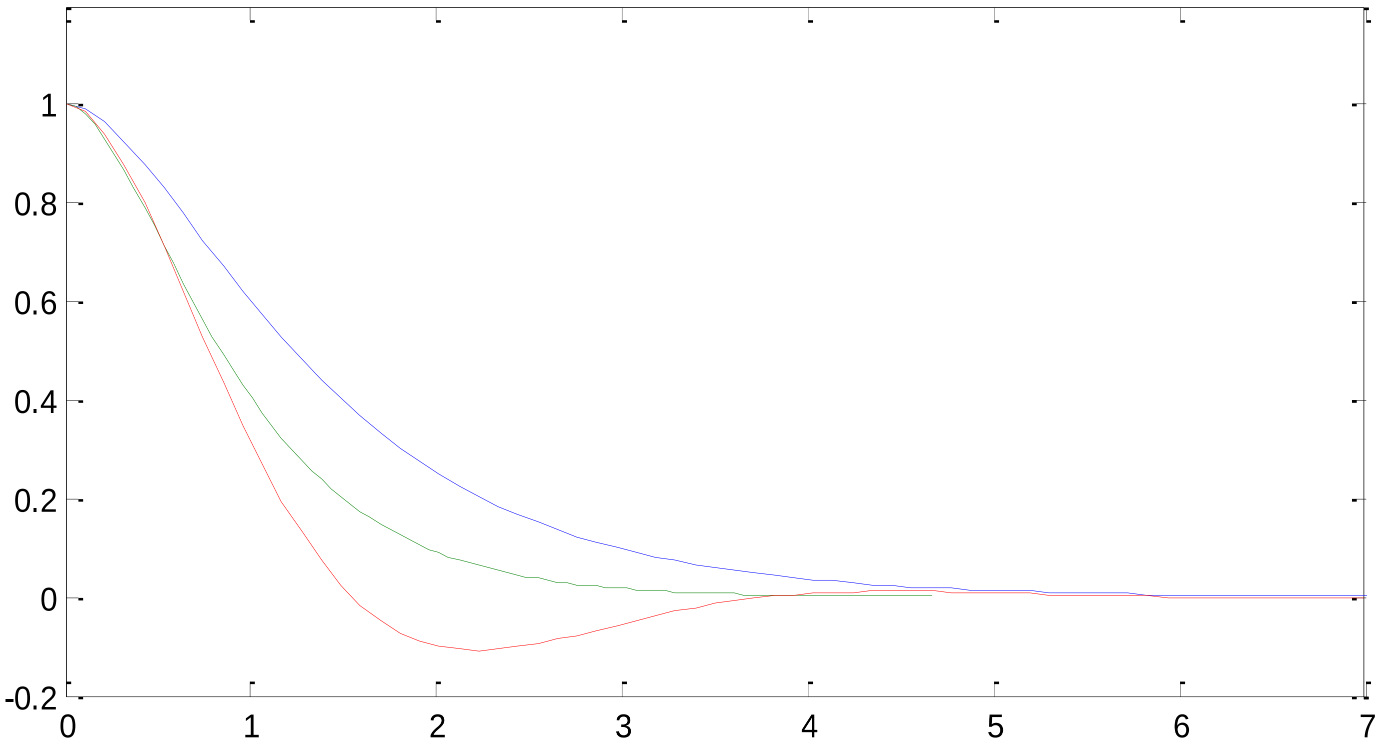
$$s_1 = -1 + j\sqrt{2}, \qquad s_1^* = -1 - j\sqrt{2}$$

$$K_1 = a + jb, \qquad K_1^* = a - jb$$

$$K_1 = a + jb = \left.\frac{P(s)}{Q'(s)}\right|_{s=s_1} = \left.\frac{s+2}{2s+2}\right|_{s=-1+j\sqrt{2}} = 0.5 - j0.3536$$

$$\frac{a+jb}{(s+\alpha)-j\omega} + \frac{a-jb}{(s+\alpha)+j\omega} = \frac{2a(s+\alpha)-2b\omega}{(s+\alpha)^2+\omega^2} \qquad 2a = 1, 2b = 0.7071$$

Primer – uporedni prikaz



Primer - nastavak

$m\ddot{x} + c\dot{x} + kx = f(t)$

Npr. za $m = 1, c = 3, k = 2, f(t) = 4, x(0) = \dot{x}(0) = 0$

Primena Laplasove transformacije na diferencijalnu jednačinu (uz nulte početne uslove) daje:

$$s^2X(s) \ + 3sX(s) \ + 2X(s) = F(s)$$

$$f(t) = 4, F(s) = \frac{4}{s}$$

$$(s^2 + 3s + 2)X(s) = \frac{4}{s}$$

$$X(s) = \frac{4}{s(s+1)(s+2)} = \frac{2}{s} + \frac{-4}{s+1} + \frac{2}{s+2}$$

$$x(t) = 2 - 4e^{-t} + 2e^{-2t}$$

Primer - nastavak

$$m\frac{d^2x(t)}{dt^2} + c\frac{dx(t)}{dt} + k \cdot x(t) = f(t)$$

$$f(t) = 10\cos(2t), \quad x(0) = x_0 = 1, \quad \frac{dx}{dt}(0) = 0,\ m = 10,\ c = 5,\ k = 0.5$$

$$m\left(s^2X(s) - sx(0) - \frac{dx}{dt}(0)\right) + c(sX(s) - x(0)) + k \cdot X(s) = F(s)$$

$$10(s^2X(s) - s) + 5(sX(s) - 1) + 0.5 \cdot X(s) = 10\frac{s}{s^2+2^2}$$

$$s^2X(s) - s + 0.5sX(s) - 0.5 + 0.05 \cdot X(s) = \frac{s}{s^2+4}$$

$$(s^2 + 0.5s + 0.05)X(s) = \frac{s}{s^2+4} + s + 0.5 = \frac{s+(s+0.5)(s^2+4)}{s^2+4}$$

$$X(s) = \frac{s^3+0.5s^2+5s+2}{(s^2+4)(s^2+0.5s+0.05)} = \frac{s^3+0.5s^2+5s+2}{s^4+0.5s^3+4.05s^2+2s+0.2}$$

$$X(s) = \frac{s^3+0.5s^2+5s+2}{(s^2+4)(s^2+0.5s+0.05)} = \frac{-0.2263}{s+0.3618} + \frac{1.4642}{s+0.1382} + \frac{-0.2379s+0.1205}{s^2+4}$$

$$x(t) = -0.2263e^{-0.3618t} + 1.4642e^{-0.1382t} - 0.2379\cos(2t) + 0.0602\sin(2t)$$

Odziv modela 2. reda

- Uvođenjem smena model se može napisati kao $\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2 x = f_{sp}(t)$

$$\zeta = \frac{c}{2\sqrt{mk}}$$ ← Faktor relativnog prigušenja

$$\omega_0 = \sqrt{\frac{k}{m}}$$ ← Prirodna učestanost

- Na karakter odziv sistema utiču polovi sistema
  - Imenilac $X(s)$ (ranije posmatrana karakteristična jednačina): $s^2 + 2\zeta\omega_0 s + \omega_0^2 = 0$ ima korene:

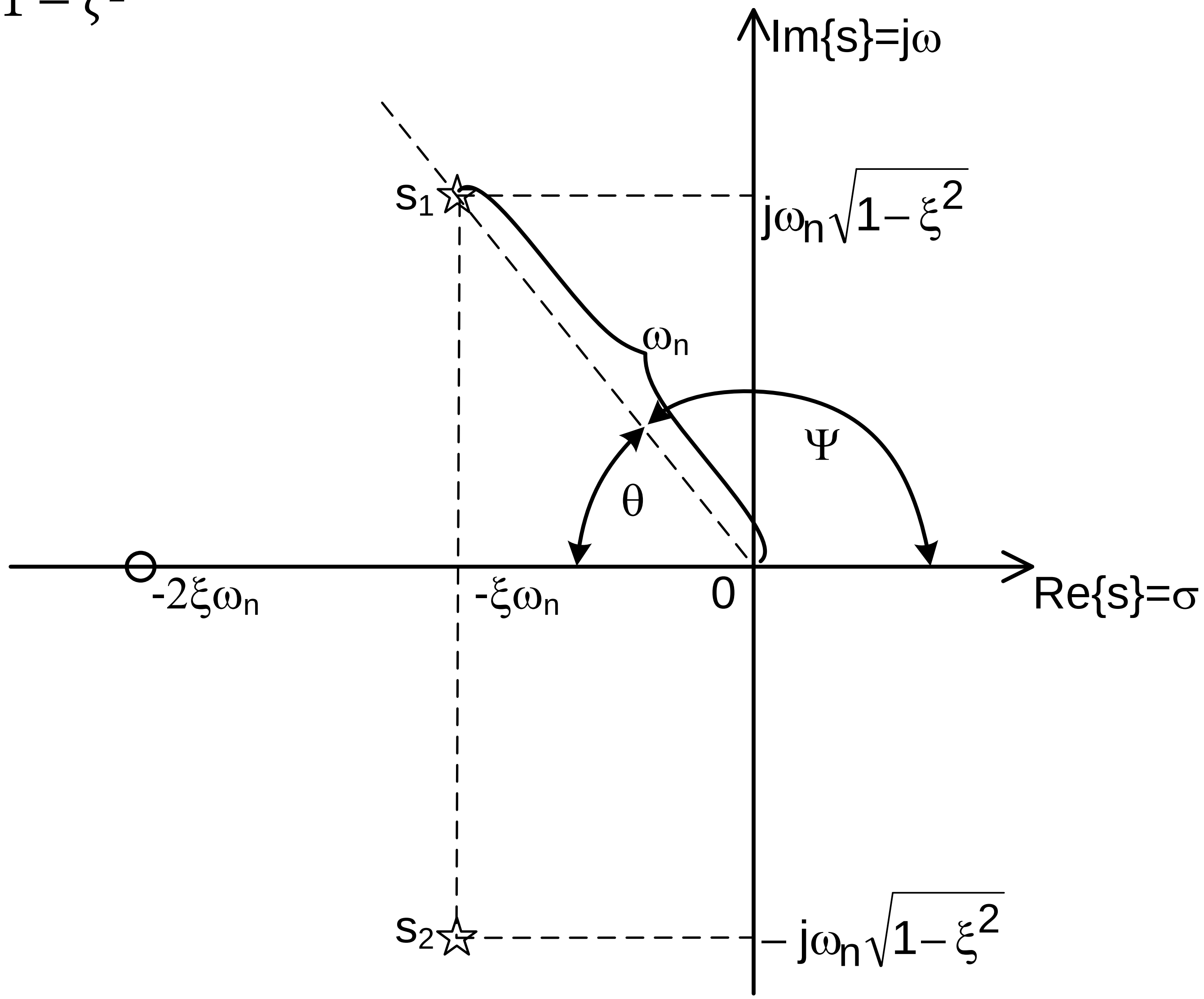
$$s_{1,2} = -\zeta\omega_0 \mp \omega_0\sqrt{\zeta^2 - 1}$$ ← Realni koreni $\zeta \geq 1$

$$s_{1,2} = -\zeta\omega_0 \mp j\omega_0\sqrt{1 - \zeta^2}$$ ← Konjugovano-kompleksni polovi $\zeta < 1$

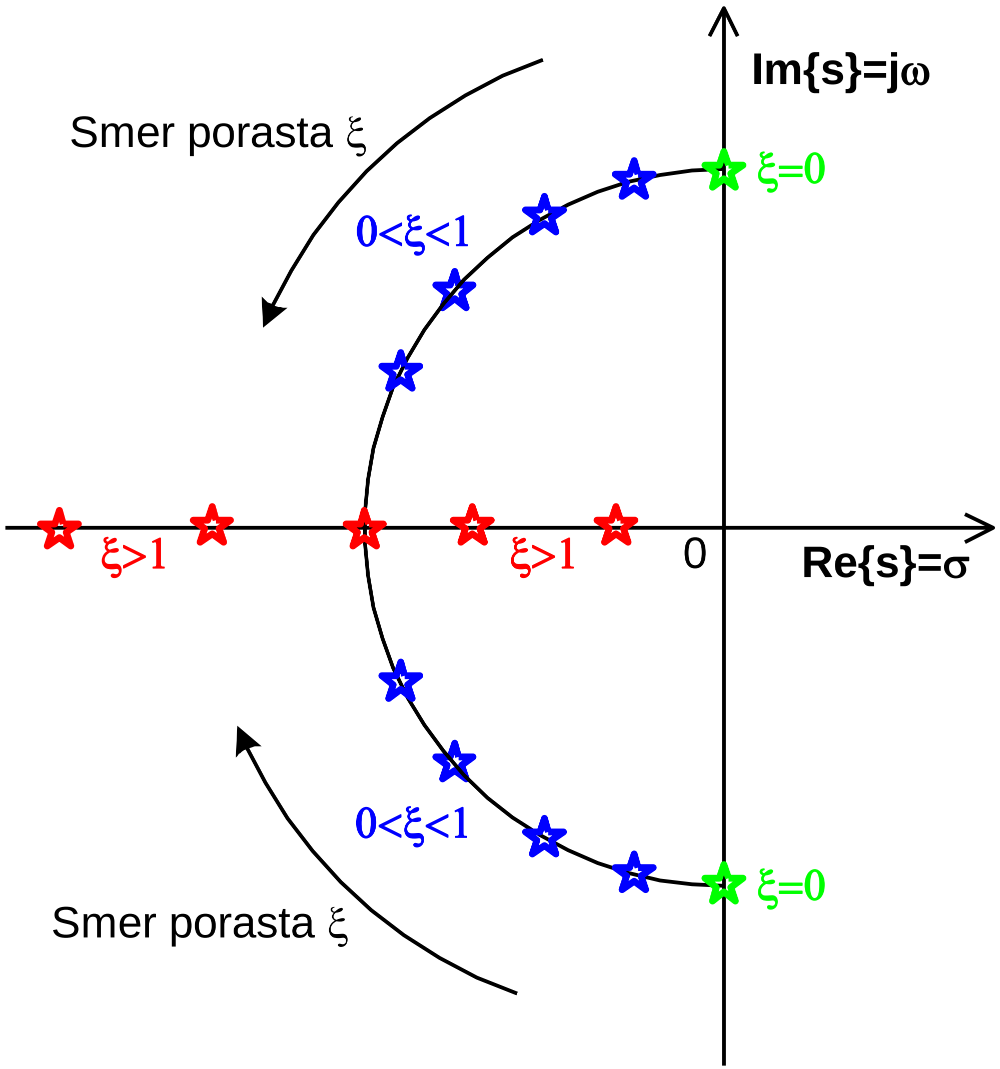
Lokacije konjugovano-kompleksnih polova

$s_{1,2} = -\zeta\omega_0 \mp j\omega_0\sqrt{1-\zeta^2}$

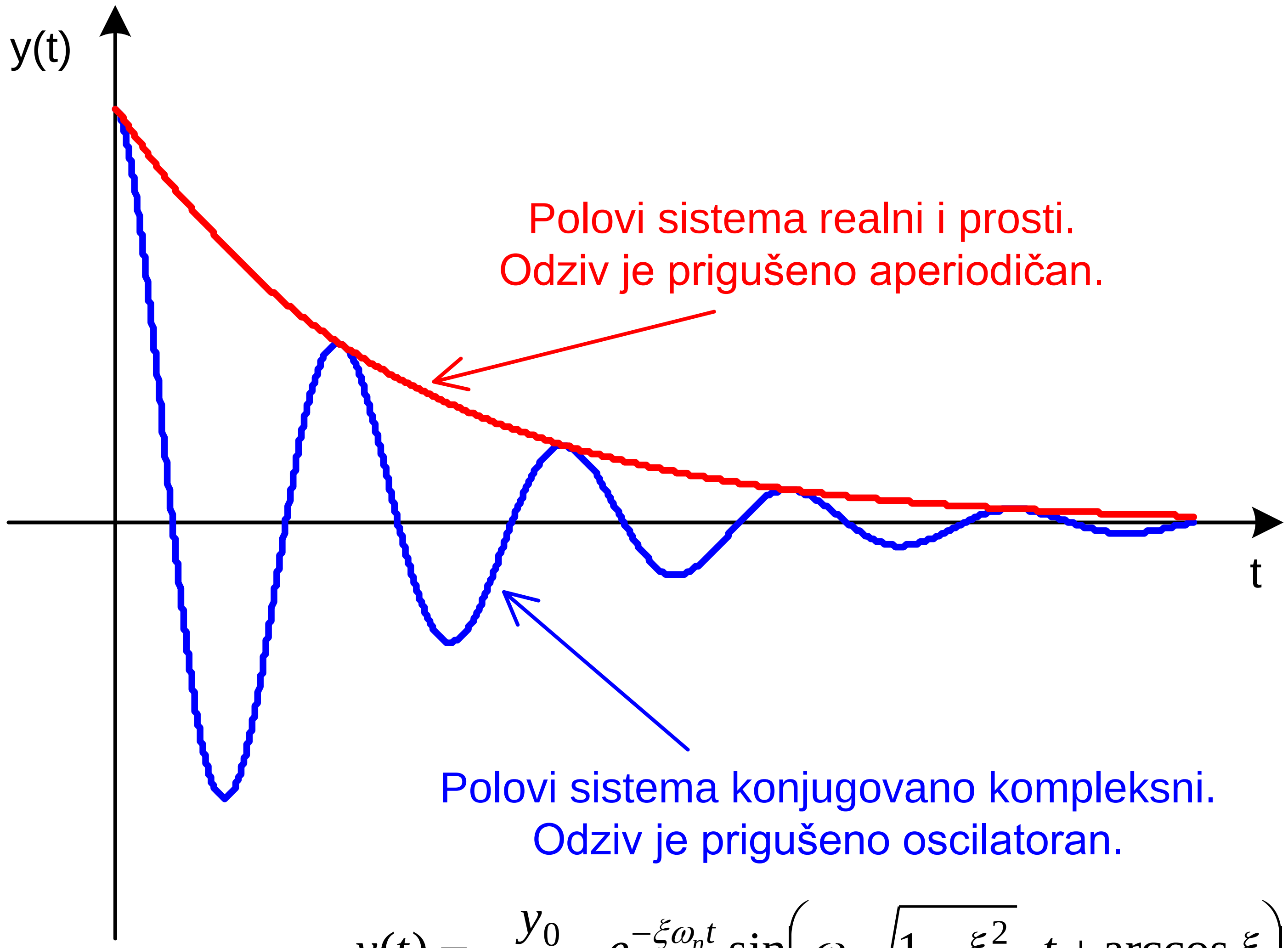
Na slici je $\omega_n = \omega_0$



Uticaj ξ na lokacije polova



Prigušen odziv sistema



$$y(t) = \frac{y_0}{1-\xi^2} e^{-\xi\omega_n t} \sin\left(\omega_n \sqrt{1-\xi^2} \cdot t + \arccos\xi\right)$$

Primer

- Poznato

$$\dddot{x}(t) + 3\ddot{x}(t) + 5\dot{x}(t) + 3x(t) = 3 + 2\sin 2t$$
$$x(0) = \dot{x}(0) = \ddot{x}(0) = 0$$

- Rešenje

$$f(t) = 3 + 2\sin 2t$$
$$\dddot{x}(t) + 3\ddot{x}(t) + 5\dot{x}(t) + 3x(t) = f(t)$$

$$s^3X(s) + 3s^2X(s) + 5sX(s) + 3X(s) = F(s)$$
$$X(s) = \frac{1}{s^3 + 3s^2 + 5s + 3}F(s)$$
$$F(s) = \frac{3}{s} + \frac{4}{s^2 + 2^2}$$

…

$$X(s) = \frac{3s^2 + 4s + 12}{s^6 + 3s^5 + 9s^4 + 15s^3 + 20s^2 + 12s}$$

$$X(s) = \frac{r_1}{s - p_1} + \frac{r_2}{s - p_2} + \frac{r_3}{s - p_3} + \frac{r_4}{s - p_4} + \frac{r_5}{s - p_5} + \frac{r_6}{s - p_6}$$

$$r_1 = -0.0235 \; + \; 0.1059j, p_1 = 2j$$
$$r_2 = r_1^*, p_2 = p_1^*$$
$$r_3 = 0.0735 \; + \; 0.1872j, p_3 = -1 \; + \; j\sqrt{2}$$
$$r_4 = r_3^*, p_4 = p_3^*$$
$$r_5 = -1.1, p_5 = -1$$
$$r_6 = 1, p_6 = 0$$

$$X(s) = \frac{-0.0471s - 0.4235}{s^2 + 4} + \frac{0.1471s - 0.3824}{s^2 + 2s + 3} + \frac{-1.1}{s + 1} + \frac{1}{s}$$

$$X(s) = -0.0471\frac{s}{s^2+2^2} - \frac{0.4235}{2}\frac{2}{s^2+2^2} +$$
$$0.1471\frac{s+1}{(s+1)^2+\left(\sqrt{2}\right)^2} + \frac{0.1471-0.3824}{\sqrt{2}}\frac{\sqrt{2}}{(s+1)^2+\left(\sqrt{2}\right)^2} + \frac{-1.1}{s+1} + \frac{1}{s}$$

$$x(t) = \mathcal{L}^{-1}\{X(s)\}$$

$$x(t) = -0.0471\cos 2t - 0.2117\sin 2t +$$
$$0.1471e^{-t}\cos t\sqrt{2} - 0.1664e^{-t}\sin t\sqrt{2} - 1.1e^{-t} + 1$$